

LABOR SERVICES EXCHANGE

The goal is to develop an application for a non-monetary exchanges of labor services within a predefined set (non-empty) of activity categories. The application accommodates a type of user called the configurator. The configurator must do the following:

1. Authentication:

- Configurators log in for the first time using default credentials(**username:** admin, **password:** admin). The default credentials are provided to each new configurator.
- During the first session, configurators must choose personal credentials for future logins these credentials must be saved in an XML file.
- Only after choosing personal credentials can they access the backend of the application.
- Each configurator's username is unique.

2. Introduce a new hierarchy of categories(look at page 3)

The configurator is responsible for defining the categories of exchangeable activities.

- Each category should consist of one or more subcategories, each corresponding to a value of its characteristic field.
- Subcategories can be recursively subdivided, forming a tree-like hierarchy.
- Leaf categories do not possess any fields.
- The same application can accommodate multiple hierarchies. There is a constraint of uniqueness for the name of each root category across all root categories within these hierarchies.

3. Introduce geographic areas where the services are available.

The configurator must configure a geographical containing all districts of the city. Each district is composed by the belonging neighborhoods. The services can be exchanged if they belong to the same district .

The districts are: Centro, Est, Nord, Ovest, Sud. The belonging neighborhoods are displayed on the following table.

NEIGHBORHOODS	DISTRICT
Brescia Antica	Centro
Centro Storico Nord	Centro
Centro Storico Sud	Centro
Crocefissa di Rosa	Centro
Porta Milano	Centro
Porta Venezia	Centro
San Polo	Est
Buffalora	Est
Caionvico	Est
San Polo Case	Est
San Polo Cimabue	Est
San Polo Parco	Est
Sanpolino	Est
Sant'Eufemia della Fonte	Est
Borgo Trento	Nord
Casazza	Nord
Mompiano	Nord
San Bartolomeo	Nord
San Rocchino	Nord
Sant'Eustacchio	Nord
Villaggio Prealpino	Nord

Chiusure	Ovest
Fiumicello	Ovest
Primo Maggio	Ovest
Urago Mella	Ovest
Villaggio Badia	Ovest
Villaggio Violino	Ovest
Chiesanuova	Sud
Don Bosco	Sud
Folzano	Sud
Fornaci	Sud
Lamarmora	Sud
Porta Cremona-Volta	Sud
Villaggio Sereno	Sud

4. Set conversion factors between leaves.

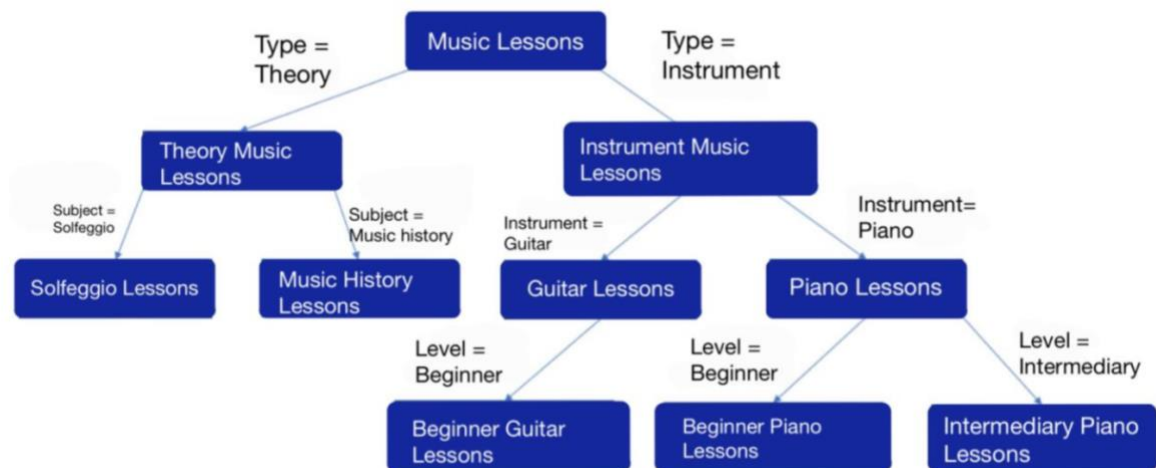
Suppose we have two leaves called c1 and c2. For the ordered pair (c1, c2) belonging either to the same hierarchy or different hierarchies, a conversion factor is established. This factor represents the value by which to multiply a quantity of labor hours related to c1 to obtain the corresponding quantity of labor hours related to c2.

- For each ordered pair of leaf categories, the conversion factor takes a real value between 0.5 and 2.0.
- If **f₁₂** is the value of the conversion factor for the ordered pair (c1, c2), then the value of the conversion factor for the ordered pair (c2, c1) is **1/f₁₂**.
- The configurator assigns manually all the necessary conversion factors and the system calculates the rest. For example:
Considering for example another leaf c3. If f₁₂ is the value of the conversion factor for the ordered pair (c1, c2) and f₂₃ is the value of the conversion factor for the ordered pair (c2, c3), then the value of the conversion factor for the ordered pair (c1, c3) is the **product** of f₁₂ and f₂₃.
- When the configurator he has already configured some preexisting hierarchies(which means that the conversion factors for all the preexisting pairs are already calculated) and he inserts a new hierarchy (lets call it G1). To calculate the conversion factor of the new pairs containing a leaf from G1 the configurator assigns one first conversion factor then the system assignments values to other conversion factors using the rule of the previous point.

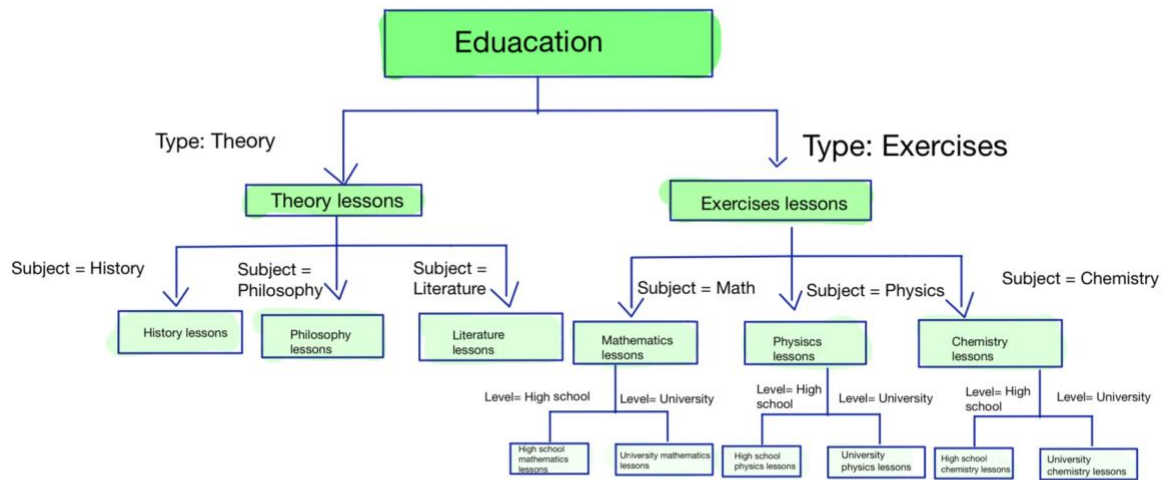
The hierarchies that need to be configured:

➤ Music

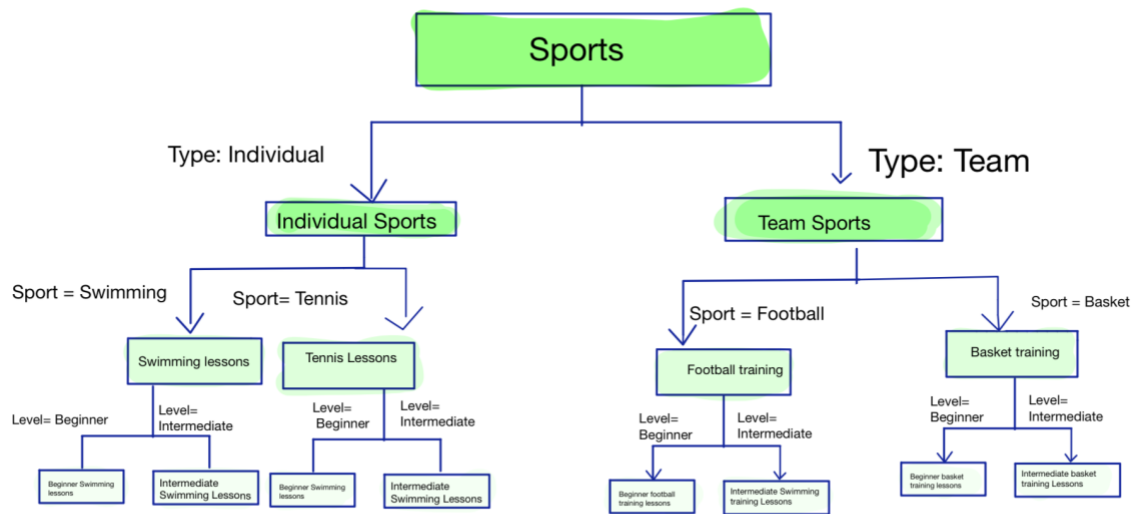
The root "Music Lessons," has the field "type," with a domain of {theory, instrument}. It branches into subcategories: "Theory Music Lessons," corresponding to the first value of the "type" field, and "Instrument Music Lessons," corresponding to the second value of the same field. The "Theory Music Lessons" subcategory features the field "subject," with a domain of {solfeggio, music history}; its subcategories are "Solfeggio Lessons," corresponding to the first value of the "subject" field, and "Music History Lessons," corresponding to the second value of the same field. Both of these are leaf categories.



➤ Education



➤ Sports



IMPORTANT NOTE

1. It is required to respect the developing patterns like GoF, SOLID and GRASP.
2. The program must allow for the possibility of adding other types of users in the future using iterative/incremental developing processes.